基本信息

姓 名：	曾海琳	籍 贯：	广东广州
性 别：	女	微 信：	_Nuyoah515
邮 箱：	h.zeng861@yahoo.com	电 话：	+86-13316083655
	353952922@qq.com		+1-4074617683

GitHub：https://github.com/hhhlin5

领英 LinkedIn：www.linkedin.com/in/hailin-zeng

个人网站：https://hhhlin5.github.io/Hailin-Portfolio/index.html

教育背景

美国佛罗里达大学 University of Florida（2026 QS：212） 2025.01 — 2026.05

计算机科学硕士学位 GPA: 3.66 / 4.00

美国佛罗里达大学 University of Florida（2026 QS：212） 2022.05 — 2024.12

计算机科学学士学位 GPA: 3.66 / 4.00

专业技能

编程语言:	C++、HTML/CSS、Python、JavaScript、TypeScript、R、MATLAB、Java、C
前端与跨端开发:	React、React Native、Vue、Angular、Expo
后端开发:	Go / Golang、Node.js、Flask、PyTorch
数据库:	PostgreSQL、MySQL、MongoDB、SQLite
DevOps & 云:	Git / GitHub、AWS、Linux / Unix、Docker
开发工具与协作:	Android Studio、Figma、Jupyter Notebook
相关课程:	数据结构与算法、高级数据结构、算法设计、分布式操作系统、计算机网络、数据库系统、软件工程、机器学习工程、自然语言处理应用、物联网安全与隐私

项目经历

Android 旅行规划应用程序 2024.9–2024.12

TripTrails(跨平台移动应用) -- 前端设计与开发工程师 | TypeScript, HTML/CSS, UI&UX, React Native, Android Studio, Expo, Git/Github

- 与团队成员协作，运用 **Figma** 将 **UI/UX** 线框图和原型转化为功能完整的跨平台移动应用。使用 **React Native** 开发用户界面前端，运用 **TypeScript** 进行实现，确保设计保真度与优质用户体验。
- **集成Google Maps API**，为用户提供交互式地图视图、位置推荐及路线规划功能。实现日历仪表盘以简化日程安排，并支持**iCal格式**实现日程共享。显著提升了用户旅行规划的便捷性与实用性，优化了日程管理的用户体验。
- 设计了简洁直观的日历仪表盘，用户可轻松查看、添加、删除、编辑及分享日程，确保用户能够快速上手并使用核心功能。
- 采用Expo构建全栈移动解决方案，管理项目配置并简化构建流程。以**Android Studio**作为核心IDE进行高级调试、性能分析及模拟器管理，显著提升开发效率并优化资源利用率。
- 在开发过程中，应用了**ESLint和Prettier**保持代码质量一致性，并使用**Jest**进行单元测试，确保代码风格一致性，稳定性和可维护性，减少了团队协作中的沟通成本。同时确保应用在不同用户工作流中的功能完整性、健壮性和可靠性。

Swimmy Gator 2024.9–2024.12

2D 游戏--Java 游戏开发工程师 | Java, OOP, UI 交互, 基础游戏物理

- 与引擎框架对接，**采用面向对象设计（OOP）** 将角色、障碍物、场景/视图等模块化封装，提升代码可维护性与迭代效率，交付可玩版本。

- 构建**多视图 UI** （开始界面/游戏中界面/结束弹窗），实现点击按钮启动游戏使界面无缝并流畅跳转，增强用户的游戏体验。
- 基于重力和冲量的物理模型实现角色运动，结合游戏循环更新与键盘事件监听，实现障碍物动态生成与行为控制，玩家通过按键盘中的 SPACE 使游戏角色想上跳动避开障碍物，随着计时器时间增加背景滚动速度越快，给玩家带来刺激感，同时完成核心操作手感调优，带来直观且具挑战性的节奏体验。
- 实现滚动背景与障碍物位移同步的视觉一致性处理，减少运动割裂与元素漂移违和感，提升画面连贯性与沉浸感。

DriveSafeUSA

2024.01 — 2024.05

数据库开发工程师 | Python, Jupyter Notebook, Oracle SQL, Node.js, Express.js, Chart.js

- 使用 **Jupyter Notebook** 中的 **Python** 脚本对 770 万条汽车事故记录进行预处理和分析（数据来自 https://www.kaggle.com）处理缺失值、标准化格式、识别关键趋势和模式（天气、碰撞、交叉、路口、车站、停车、交通），将清洗后的高质量数据高效批量导入 **Oracle** 数据库。
- 基于分析需求，设计了 **schema 数据库结构**，使用大量**数据操纵语言(DML)**，**数据定义语言(DDL)**和**数据控制语言(DCL)**，编写并优化了 **Oracle SQL** 查询。提升复杂查询性能，使宏观趋势查询的平均响应时间控制在 3s 以内。
- 使用 **Node.js** 与 **Express.js** 框架开发了安全的反向代理服务器，以促进前端与数据库之间的通信。并对所有用户输入进行严格的验证和清理，实现了API速率限制。从而降低数据泄露，防止跨站脚本攻击和SQL注入攻击等一系列网络攻击的风险。
- 实现了一个动态用户界面，通过交互式图表按时间顺序显示数据。用户可通过时间轴选择器和条件筛选器无缝切换查询，实时对比不同数据集。基于**Chart.js** 绘制时序折线图，并实现了缩放和数据点提示功能，让用户既能纵览宏观趋势，又能放大查看任何时间点的详细数据。

爬虫操作系统核心组件研发

2024.01 — 2024.05

课程项目—DevOps 工程师 | C, C++, FUSE, Git

- 设计并封装了 **log_write(level, message)** 核心API，通过操作系统内设备文件将不同严重级别的日志条目直接写入内核环形缓冲区。使用 CMake 构建系统生成静态库。从而实现与内核日志级别**交互的C API**，增强了分布式环境下的系统可调试性。
- 开发了一个**C++内存管理器**，通过维护一个红黑树数据结构来记录所有分配块。支持内存的跟踪、分配与释放，并通过大量内存泄漏和错误测试验证其正确性。优化了内存碎片处理，实现了内存块的合并与高效利用，显著提升了系统性能。
- 基于**FUSE API**开发了用户空间文件系统守护进程，实现了 **getattr, read, readdir** 等核心文件操作回调函数。实现了与**WAD文件的交互，**通过解析WAD中的的文件和目录结构，将其映射为**FUSE虚拟文件系统**的树状结构，支持用户对WAD文件内文件与目录通过**Shell命令**进行读写访问，成功演示了文件系统操作的概念验证功能，验证了设计的可行性与实用性。

智能综合评分系统

2023.10 — 2023.12

全栈开发工程师 | Full-Stack Web development, Security and Role-Based Access Control, JavaScript, C++, HTML/CSS, MongoDB, Docker

- 基于 **C++** 开发了高性能的静态代码分析脚本，通过集成 **LLVM/Clang** 对提交的源代码进行编译检查、语法分析和基础逻辑验证。将原本需手动逐行检查的评分流程自动化，单次作业批改时间**从平均10分钟缩短至5分钟以内**。
- 使用**React.js**与**Chart.js**框架构建了并实现了成绩分布可视化图表、实时成绩更新、以及友好的代码提交反馈界面。通过**Axios**与后端进行异步数据交互，页面响应迅速以及数据呈现直观友好，**用户满意度调查显示评分提升25%**。
- 基于**Node.js**和**Express.js**框架开发设计了**RESTful API**，处理用户认证、作业提交、成绩查询等请求。优化 **MongoDB** 的文档结构设计并建立高效索引，并对常用查询接口进行数据库操作优化，确保**API平均响应时间在500ms以内**，承受住了高峰时段百人并发的压力测试。
- 实现了基于角色（RBAC）的访问控制，区分学生、教师和管理员权限。使用 **JWT** 实现无状态的身份验证，用户密码通过 **bcrypt** 算法进行**哈希函数**加密存储。严格的后端接口校验确保了用户只能访问其权限范围内的数据，经安全扫描评估，**关键数据泄露风险降低70%**。
- 将前端、后端、数据库等服务分别封装至 **Docker 容器**，实现了开发、测试、生产环境的一致性。在**AWS**云平台部署应用程序，使学生提交的代码文件自动存储至 AWS，实现了高可用性与无缝访问，确保了系统的可靠性与可扩展性。

荣誉 & 证书

2021 年美国校际程序设计竞赛	04/2021
计算机科学学士学位荣誉毕业：Magna Cum Laude（GPA 前 20%）	12/2024
PMI	IIBA	NASBA – Agile Foundations（敏捷基础）结课证书	01/2026
AWS Certified Cloud Practitioner	

自我评价

我工作积极认真、细心负责，具备强烈责任感与高效的团队协作意识与能力。做事自律有条理，能通过快速学习与独立解决问题推动任务按时高质量落地。我勤奋好学、动手能力强，乐于迎接新挑战并能吃苦耐劳，面对新环境与压力也能保持积极韧性。

Hailin Zeng

h.zeng@ufl.edu | h.zeng.861@yahoo.com | 407-461-7683 | Gainesville, FL
https://www.linkedin.com/in/hailin-zeng/ | https://github.com/hhhlin5| https://hhhlin5.github.io/Hailin-Portfolio/

Education

Master of Science in Computer Science 01/2025-05/2026
University of Florida, Gainesville, FL GPA: 3.66 / 4.00

Bachelor of Science in Computer Science 05/2022-12/2024
(Graduated with honor - Magna Cum Laude – top 20%) GPA: 3.66 / 4.00
University of Florida, Gainesville, FL

Associate in Arts (A.A.) Degree
(Graduated with honor) 08/2019-04/2022
Valencia College, Orlando, FL GPA: 3.81 / 4.00

Technical Skills

- Programming Languages: C++(proficient), Python(proficient), JavaScript(proficient), HTML/CSS(proficient), TypeScript, Java, R, MATLAB
- Frontend & Backend: React Native(proficient), React.js(proficient), Vue.js, Angular, Node.js, Flask, Golang
- DevOps & Cloud & Databases & OS: AWS, Linux/Unix, Docker, PostgreSQL, MongoDB, SQLite
- Development Tools & Platforms: Git/GitHub, Jupyter Notebook, Android Studio, Expo, Figma, HuggingFace

Experience

2021 Intercollegiate Programming Competition *(python)* 04/2021

- Performed rapid prototyping and algorithm implementation based on Python, translating theoretical algorithms into efficient and reliable code, solving complex problems involving dynamic programming, graph theory, and advanced data structures.
- Reduced the time complexity of multiple solutions from $O(n^2)$ to $O(n \log n)$ through systematic algorithm optimization, significantly improving the execution efficiency of the code with large-scale data.

Projects

Android Travel Planning App --- TripTrails *(Mobile Dev)* 09/2024 – 12/2024
Frontend Developer | TypeScript, HTML/CSS, UI&UX, React Native, Android Studio, Expo, Git/Github

- Collaborated with team members to transform **UI/UX wireframes** and prototypes in **Figma** into fully functional cross-platform mobile applications. Developed front-end user interfaces use **TypeScript and JavaScript** with **React Native**, ensuring user has premium experiences.
- Integrated **Google Maps API** to deliver interactive map views, location recommendations, and route planning capabilities. Implemented a calendar dashboard UI enabling users to effortlessly view, add, delete, edit, and enabled **iCal format** support for calendar sharing. Significantly enhanced the convenience and practicality of travel planning while optimizing the user experience for schedule management.
- Architected and developed a full-stack mobile solution using **Expo** to manage the project configuration and simplify the build process. Utilized **Android Studio** as the primary IDE for advanced debugging, performance profiling, and emulator management, significantly improving development efficiency and ensuring optimal resource utilization.
- Applied **ESLint and Prettier** for maintaining high code quality and consistent formatting standards. Utilized **Jest for unit testing**, reducing communication overhead in team collaboration, and ensuring the app's functionality, robustness, and reliability across different user workflows.

Operating System in Reptilian 01/2024 – 05/2024
Systems Engineering | C, C++, Linux/Unix, CMake, Kernel logging, Memory management, FUSE, Filesystem

- Designed and encapsulated a core log_write(level, message) API in **C**. This API writes log entries of varying severity levels directly to the kernel's ring buffer via an OS device file. Built a static library using the **CMake** build system, creating a **C API** that interfaces with kernel log levels to enhance system debuggability in distributed environments.
- Engineered a **C++ memory manager** that tracks all allocated blocks by maintaining a **Red-Black Tree**. The system supports **memory tracing, allocation, and deallocation**, with its correctness rigorously validated through extensive memory leak and error testing. Optimized memory fragmentation handling by implementing block coalescing for efficient memory utilization, significantly improving system performance
- Developed a user-space filesystem daemon using the **FUSE API**, implementing core file operation callbacks such as **getattr, read, and readdir**. Enabled **interaction with WAD files** by parsing their internal file and directory structures and mapping them to a FUSE virtual filesystem tree. Allowed users to read and access files within the WAD archive directly via standard shell commands, successfully demonstrating a proof-of-concept for the filesystem operations and validating the design's feasibility and practicality.

Integrated Grading System *(Full stack)* 10/2023 – 12/2023
Full Stack Developer | Node.js, Express.js, MongoDB, Docker, AWS (S3), React.js, Chart.js, C++ (LLVM/Clang), JWT, RBAC

- Integrated **LLVM/Clang** in a **C++ static analysis pipeline** to perform compilation checks, syntax parsing, and basic logic validation on submitted code. Automated the manual grading process, reducing the average time to grade from 10 minutes to under 5 minutes.
- Built a dynamic and responsive front-end using **React.js** and **Chart.js**, implementing features such as grade distribution visualizations, real-time score updates, and a code submission interface. Utilized **Axios** for asynchronous data exchange with the backend, ensuring rapid page responsiveness and intuitive data presentation. A user satisfaction survey indicated a 25% improvement in scoring experience.
- Built and operated a **Node.js/Express** backend with **REST APIs** for user authentication, assignment submissions, and grade inquiries. Optimized **MongoDB** document structure. Ensured API response times under 500ms and successfully withstood stress tests simulating hundreds of concurrent users during peak hours.
- Strengthened security controls for production systems with **RBAC** (student/teacher/admin). Established stateless authentication using **JWT** and securely stored user passwords with **bcrypt hashing**. Enforced server-side authorization checks to ensure users could only access data within their permissions, reducing the risk of critical data leakage by ~70%.
- Improved production reliability and release consistency by containerizing each service with **Docker**, enabling repeatable dev/test/prod environments and smoother release handoffs across teammates. Operationalized file handling in the cloud by deploying on **AWS** and storing submitted code artifacts in **S3**, reducing application server load and simplifying data retention/backup workflows.

Certifications & Licenses

AWS Certified Cloud Practitioner – Basic (pending)
PMI | IIBA | NASBA – Certificate of Agile Foundations